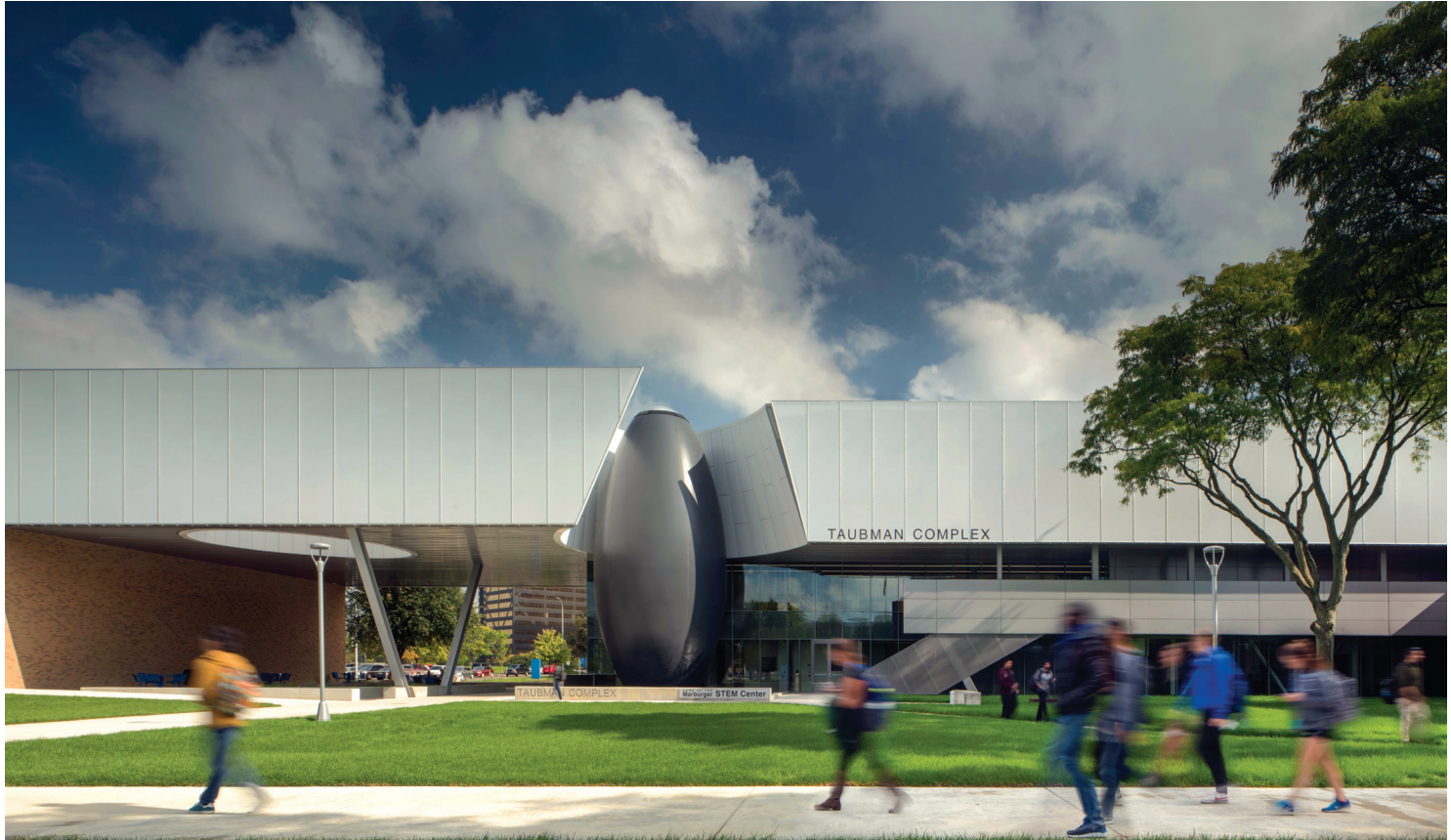




BIOMEDICAL ENGINEERING

Lawrence Technological University

Lawrence Technological University's Department of Biomedical Engineering (BME) provides a student-centered learning environment that encourages interdisciplinary research and design to solve biomedical problems. LTU's state-of-the-art laboratory spaces located in the A. Alfred Taubman Engineering, Architecture, and Life Sciences Complex offer a broad range of BME sub-disciplines to explore, such as medical devices, bioinstrumentation, biomechanics, biomaterials, and tissue engineering.

With easy access to the lab facility and faculty, BME students gain valuable hands-on skills by participating in a variety of research and design projects, which prepare them to become leaders and innovators in their future careers.



"Project-based learning activities in both classrooms and labs help in my career."

– Monica Ellicott, BSBME'19

Experimental Biomechanics

LABORATORY (EBL)

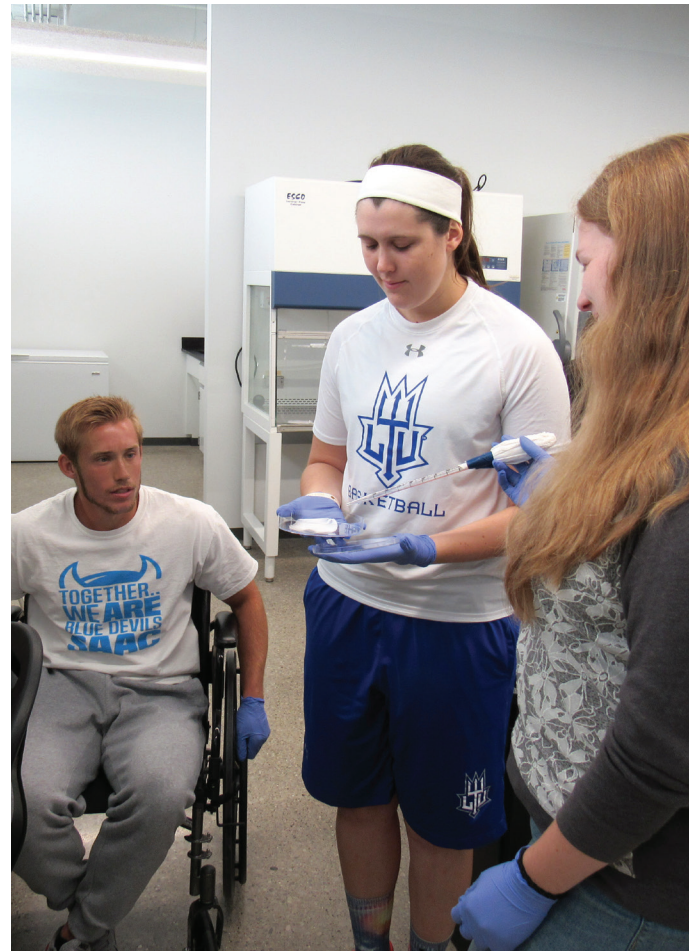
EBL advances experimental biomechanics understanding and provides practical training to engineering, life science and medical students, advancing the boundary of knowledge through translational research.

EBL is equipped with a 10 camera Vicon optical motion analysis system, two Kistler force platforms, and a Delsys® Trigno 16-channel, wireless sensor system with a variety of physiological monitoring sensors. The custom-designed lab enables advanced research on motion of the human body, as well as emerging areas of robotics, haptics, and development of wearable technology devices.



Cell Culture and Cell Biology LABORATORY

LTU's Cell Culture Lab is shared by students and faculty in BME and Natural Sciences. It houses six cell culture hoods, a spectrophotometer, phase and fluorescence microscopes, and other small equipment for cell culturing and characterization. The newly added bioprinter enables exciting cell and tissue engineering research.



Biomaterials LABORATORY

LTU's Biomaterials Lab provides equipment and supplies for students and faculty to synthesize and process different biomaterials for orthopedic, drug delivery and tissue engineering applications. The lab also has an FT-IR spectrometer and rheometer for structure analysis of biomaterials, and a variety of surgical and dental devices and implants.

"LTU provided the foundation for out-of-the-box thinking."

– Dori Watts, BSBME'18



Bioinstrumentation LABORATORY

LTU's Bioinstrumentation Lab allows the measurement and utilization of signals from the heart (ECG), brain (EEG), and muscles (EMG) for diagnostics and innovative therapies. It houses a variety of electronic equipment and supplies to enable the design, fabrication and testing of bioelectronic and wearable devices.



Environmental Scanning Electron Microscope (ESEM) LABORATORY

LTU's ESEM Lab is a National Science Foundation funded lab that allows users to conduct microscopic observation of any materials and structures at the nanoscale resolution. The microscope is also equipped with an energy dispersive detector for elemental analysis, a hot and cold stage, and a microtester for in situ mechanical testing of materials.



BioMEMS and Microfabrication LABORATORY

LTU's BioMEMS and Microfabrication Lab enables the fabrication of miniaturized devices to revolutionize biomedical investigation and clinical practice. The lab has a Class 1000 cleanroom with photolithography and wet etching capability; and a polymer MEMS area that focuses on fabricating and testing of polydimethylsiloxane-based microdevices using soft lithography.

Biofabrication LABORATORY

Students create prototypes of their engineering designs in LTU's Biofabrication Lab. It houses a variety of tools, such as 3D printers (fused filament-based and resin-based), a CNC machine, and a laser cutter.

"When you take senior projects you will learn how you like to work and that will begin to impact career decisions."

— Kevin Mazure, BSBME'19

Lawrence Technological University

, founded in 1932, offers more than 100 programs through the doctoral level in the Colleges of Architecture and Design, Arts and Sciences, Business and Information Technology, and Engineering. Activities on Lawrence Tech's 107-acre campus include more than 100 student organizations and NAIA varsity sports. LTU produces leaders with an entrepreneurial spirit and a global view. Co-ops and internships with key industry partners abound. That's why 84% of Lawrence Tech students are employed or registered for grad school at commencement and Payscale ranks the earning power of a Lawrence Tech bachelor's degree in the top 11% of all U.S. universities.

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"I still use many of the problem-solving skills I learned in the lab."

– Nicole Buckingham, BSBME'19



COLLEGE OF ENGINEERING
**BIOMEDICAL
ENGINEERING**

21000 West Ten Mile Road
Southfield, MI, 48075-1058

LTU's Department of Biomedical Engineering offers a bachelor's degree in biomedical engineering and a master's degree in biomedical engineering and cardiovascular perfusion. LTU's bachelor's degree in biomedical engineering is ABET-accredited.

Dedicated faculty and staff provide students with a personalized education, including extensive laboratory work that emphasizes "Theory and Practice."

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